

Day 27 – Confidence & Sentiment Signal Analysis

Objective

To assess communication quality and behavioral indicators during the AI screening process.

Detailed Tasks

1. Detect Hesitation Patterns

Analyze candidate responses for hesitation signals such as:

- "um"
- "uh"
- "hmm"
- "maybe"
- "I think"
- "probably"
- Repeated words
- Long pauses or hesitation indicators

These signals help identify uncertainty or lack of confidence in communication.

2. Measure Response Length and Pace

Analyze the candidate's response characteristics, including:

- Number of words
- Response length
- Number of sentences
- Speaking/response pace when timing information is available
- Very short responses
- Excessively long responses

These indicators help evaluate communication efficiency and clarity.

3. Identify Positive/Negative Sentiment

Analyze the candidate's response to identify the overall sentiment.

Possible sentiment categories:

Positive

Neutral

Negative

Positive responses may indicate confidence, enthusiasm, and constructive communication.

Negative responses may indicate dissatisfaction, uncertainty, frustration, or other communication concerns.

4. Detect Uncertainty and Contradictions

Identify uncertainty indicators such as:

- "I don't know"
- "I'm not sure"
- "maybe"
- "probably"
- "I think"

The system should also identify contradictory information between responses where possible.

Example:

Response 1:

"I have 3 years of Python experience."

Response 2:

"I started learning Python last year."

This may be flagged as a potential contradiction for recruiter review.

5. Create Communication Strength Indicators

The system should convert communication signals into meaningful indicators.

Example indicators:

Confidence Level

Response Quality

Communication Clarity

Sentiment

Hesitation Level

Uncertainty Level

Consistency

Response Length

These indicators can help recruiters understand the candidate's communication behavior.

Deliverables

1. Confidence Analysis Logic

The confidence analysis component evaluates communication signals such as:

- Confidence
- Hesitation
- Uncertainty
- Response quality
- Communication strength

Status: Completed 

2. Sentiment Scoring Module

The sentiment scoring module will analyze candidate responses and classify their sentiment as:

Positive

Neutral

Negative

It will provide a sentiment score that can be used by the screening system.

Status: To be implemented 

3. Behavioral Indicators Report

The behavioral indicators report will combine communication signals into a recruiter-friendly result.

Example:

Candidate ID: CAND-001

Confidence: High

Sentiment: Positive

Hesitation: Low

Uncertainty: Low

Communication Clarity: High

Response Length: Good

Consistency: High

Overall Communication Strength:

Strong

Status: To be implemented ⌚

Day 27 Processing Flow

Candidate Response



Confidence Analysis



Hesitation Detection



Response Length & Pace Analysis



Sentiment Analysis



Uncertainty Detection



Contradiction Detection



Communication Strength Indicators



Behavioral Indicators Report

Expected Final Output

```
{  
  "candidate_id": "CAND-001",  
  "confidence_level": "high",  
  "sentiment": "positive",  
  "hesitation_level": "low",  
  "uncertainty_level": "low",  
  "communication_clarity": "high",  
  "response_length": "good",  
  "consistency": "high",  
  "communication_strength": "strong"
```